Arrhythmia on ECG Classification Using CNN

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I. Introduction

Cardiovascular disease is a widespread condition that poses a major threat to human health, particularly among the middle-aged and elderly.The worsening of cardiovascular disease has turned into a huge public health issue. The electrocardiogram (ECG) is an effective method that is widely used to monitor cardiac health.Cardiovascular diseases require the identification and classification of ECG signals. The electrocardiogram (ECG) is a real-time visual time series that records the electrical activity generated by each cardiac cycle of the heart.Cardiologists and medical practitioners use it extensively to monitor cardiac health.The ability of a cardiologist and medical practitioner to diagnose cardiovascular illness is largely dependent on his or her experience.Unfortunately, manually evaluating a significant volume of ECG data for a person takes a long time and is prone to errors.How to swiftly and accurately analyze certain diseases is a new dilemma that has arisen.Furthermore, the characteristics of ECG signals include unpredictable, low-frequency, and vulnerable, leading in unreliable diagnosis outcomes. To increase the efficiency and accuracy of ECG recognition, intelligent automatic recognition and classification of ECG signals has become an unavoidable option.

In this project, we used convolutional neural network (CNN) to classify heartbeats which is able to accurately classify five different arrhythmias.

II. Related Works

With the advancement of Artificial Intelligence (AI) technology, several machine learning approaches are being applied in the detection of ECG signal features, with the goal of resolving issues such as enormous volumes of ECG signal feature data and a heavy manual detection burden.The common methods are neural networks (NN) proposed by Jiang, et al [1], support vector machine (SVM) [2], path forest [3], Independent Component Correlation Algorithm (ICA) [4].Jiang, et al [1] proposes an evolutionary block-based neural network (BbNNs) for the classification of ECG data in terms of neural networks. The BbNN is made up of a collection of two-dimensional modular networks with a configurable internal structure and configuration.

In recent years, the machine learning and deep learning network has not only made remarkable achievements in the fields of image processing, audio recognition and many other fields.Zubair, et al [5] has proposed a method where it is used in the assisted diagnosis of heart disease based on ECG signals.

III. Project Objectives

The objective of our project is to classify five different arrhythmias. For this, we divide the total task into few subtasks. These are:

A. Data Pre-processing

class 1	class 2	class 3	class 4	class 5
72471	6431	5788	2223	641

From the training dataset we can see that one class has maximum number of data, which might result the model from inclining towards the majority class. We used downsampling in this case. Downsampling is the process of reducing the sampling rate of a signal. Downsample reduces the sampling rate of the inputs by an integer factor by picking up one out of N samples. There's sparsity of connections means that some parameters are simply missing (i.e., are zero), nothing to do with sharing same non-zero parameter. Parameters in this case are zero, ignored. Upsampling is a procedure where synthetically generated data points (corresponding to minority class) are injected into the dataset. After this process, the counts of both labels are almost the same. This equalization procedure prevents the model from inclining towards the majority class. That is why for class 2-5, we upsampled our training data. After resapmling all the classes each contained 20% of the whole training dataset.

B. Splitting the dataset

For validation purposes, we split the training dataset into training and validation.

C. Classification

Classifies (heartbeats) ECG signals into 5 different arrhythmias.

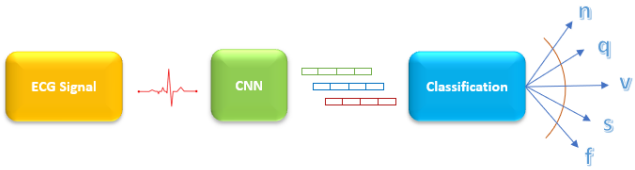

Fig. 1. Overview of our proposed system

IV. Methodologies

Our main approach to classifying the ECG beat types of the MIT-BIH dataset was to build a convolutional neural network. We proposed a network architecture for this beat classification task which is illustrated in Figure 3. As explained in pre-processing, we used the resampled data as inputs. A convolution layer was applied followed by a batch normalization layer and a max pooling layer and repeating this combination three times. Then after flattening and feeding the output data to three dense layers with two ReLU and a softmax activation function respectively, to predict output class probabilities. All the convolution layers are applying 1-D convolution through time. In total, the resulting network is a deep network consisting of 13 weight layers. For the optimizer, Adam optimizer was used as well.

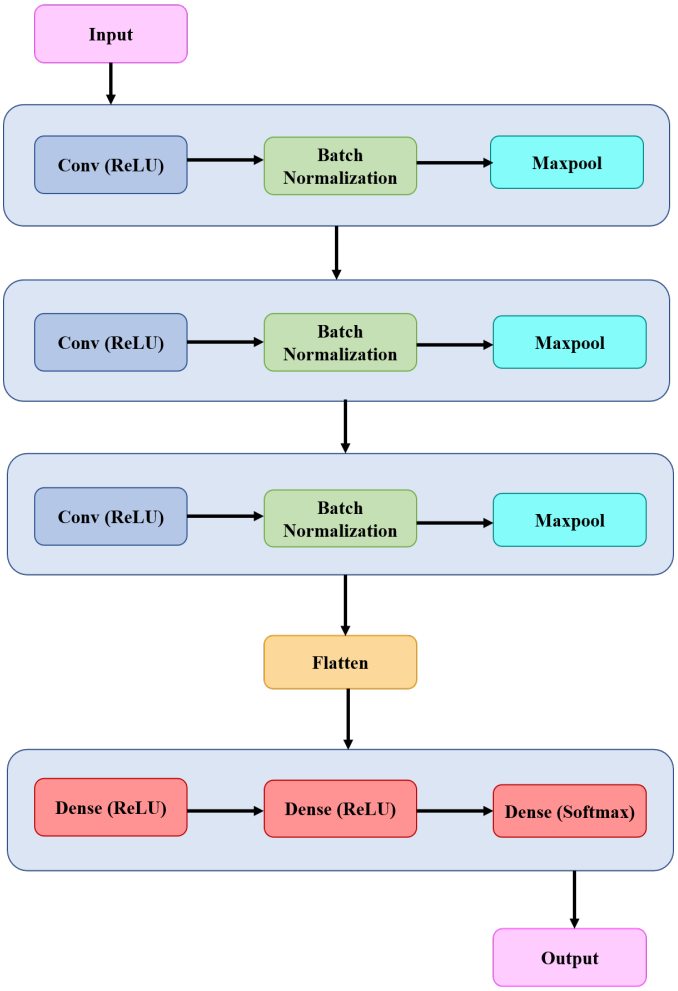

Fig. 2. Architecture of the proposed network.

```
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
inputs_cnn (InputLayer)      [(None, 186, 1)]          0
_________________________________________________________________
conv1d_3 (Conv1D)            (None, 181, 64)           448
_________________________________________________________________
batch_normalization_3 (Batch (None, 181, 64)           256
_________________________________________________________________
max_pooling1d_3 (MaxPooling1 (None, 91, 64)            0
_________________________________________________________________
conv1d_4 (Conv1D)            (None, 89, 64)            12352
_________________________________________________________________
batch_normalization_4 (Batch (None, 89, 64)            256
_________________________________________________________________
max_pooling1d_4 (MaxPooling1 (None, 45, 64)            0
_________________________________________________________________
conv1d_5 (Conv1D)            (None, 43, 64)            12352
_________________________________________________________________
batch_normalization_5 (Batch (None, 43, 64)            256
_________________________________________________________________
max_pooling1d_5 (MaxPooling1 (None, 22, 64)            0
_________________________________________________________________
flatten_1 (Flatten)          (None, 1408)              0
_________________________________________________________________
dense_2 (Dense)              (None, 64)                90176
_________________________________________________________________
dense_3 (Dense)              (None, 32)                2080
_________________________________________________________________
main_output (Dense)          (None, 5)                 165
=================================================================
Total params: 118,341
Trainable params: 117,957
Non-trainable params: 384
_________________________________________________________________
```

Fig. 3. Model for Arrhythmia on ECG Classification Using CNN

V. Experiments

A. Dataset

For our project, we used ECG Heartbeat Categorization Dataset(MIT-BIH dataset) from Kaggle. The MIT-BIH dataset consists of ECG recordings from 47 different subjects recorded at the sampling rate of 360Hz. Each beat is annotated by at least two cardiologists. The dataset contains 186 columns and 87554 rows in total. We categories the dataset into five different classes ['N': 0, 'S': 1, 'V': 2, 'F': 3, 'Q': 4] depending on unique set of values. As we can see from the visualization almost 82% (72471) of the total training dataset falls into one category. Furthermore, for validation purpose, we split the training data into training and validation with 90-10 ratio.

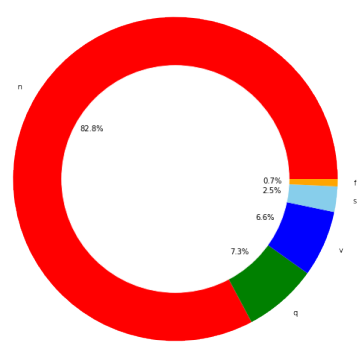

Fig. 4. Data Visualization

B. Evaluation Metric

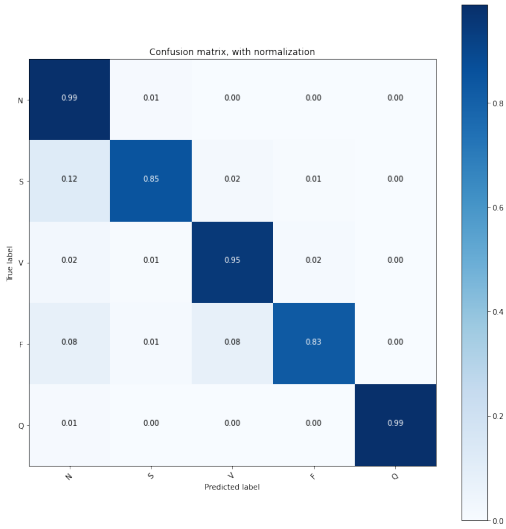

Fig. 5. Confusion Matrix

C. Results

In this section, we present the results and discuss the outcome. We show the performance of the approaches and also evaluate some of their key parameters to obtain some insights about the suitability of this approach for the domain. From the accuracy and loss graph we can see that, till 30 epochs there was drastic change in both of them but after that it was almost linear. That is why we tuned the hyperparameter to 50 epochs and batch size was 32. The result we got from that was quite satisfactory.

Class	Accuracy	Precision	Recall	F1-score
1	98%	99%	99%	99%
2		72%	85%	78%
3		95%	95%	95%
4		75%	83%	79%
5		99%	99%	99%

VI. Conclusion and Future Directions

Cardiovascular disease is a major health problem in today's world. The early diagnosis of cardiac arrhythmia highly relies on the ECG. Unfortunately, the expert level of medical resources is rare, visually identify the ECG signal is challenging and time-consuming. The proposed CNN network shows an outstanding performance in the overall classification accuracy of 98% . In our study, we used convolutional neural network (CNN) to classify heartbeats which is able to accurately classify five different arrhythmias. In terms of future study, it would be fascinating to look into using optimization techniques to come up with a feasible design and solution. Our study's limitation is that we have yet to apply any optimization techniques to improve the model parameters; however, we believe that with the implementation of the optimization, the proposed solution's performance will be able to be elevated to the next level.

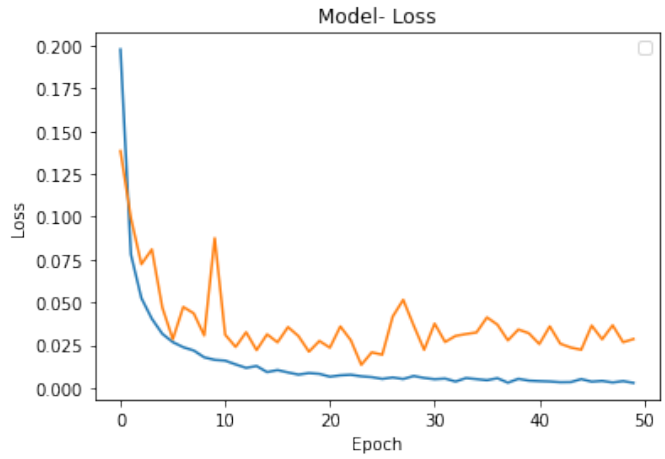

Fig. 6. Epoch vs Loss

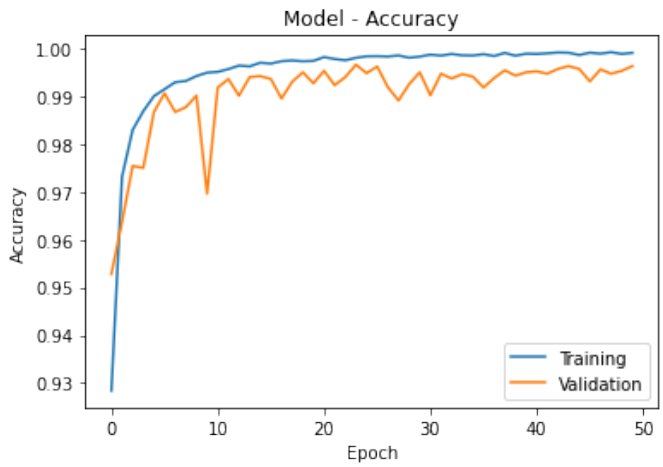

Fig. 7. Epoch vs Accuracy

References

[1] Jiang, Wei, and Seong G. Kong. "Block-based neural networks for personalized ECG signal classification." IEEE Transactions on Neural Networks 18.6 (2007): 1750-1761.

[2] Osowski, Stanislaw, Linh Tran Hoai, and Tomasz Markiewicz. "Support vector machine-based expert system for reliable heartbeat recognition." IEEE transactions on biomedical engineering 51.4 (2004): 582-589.

[3] Luz, Eduardo Jose da S., et al. "ECG arrhythmia classification based on optimum-path forest." Expert Systems with Applications 40.9 (2013): 3561-3573.

[4] Sarfraz, Mohammad, Ateeq Ahmed Khan, and Francis F. Li. "Using independent component analysis to obtain feature space for reliable ECG Arrhythmia classification." 2014 IEEE international conference on bioinformatics and biomedicine (BIBM). IEEE, 2014.

[5] Zubair, Muhammad, Jinsul Kim, and Changwoo Yoon. "An automated ECG beat classification system using convolutional neural networks." 2016 6th international conference on IT convergence and security (ICITCS). IEEE, 2016.